# Statistical Analysis of Breast Cancer Screening Uptake

## 1. Summary of Missing Data

Across all 12 clinical teams, the combined dataset comprised 528 patients. Missing data were limited to alcohol intake (37/528; 7.0%), employment status (12/528; 2.3%), and height (8/528; 1.5%) (Table 1). All other variables were complete. Overall, 479 patients (91%) had complete data across all recorded characteristics.

| Variable | Missing counts (%) |
|---|---|
| *Alcohol intake* | 37 (7.0%) |
| *Employment status* | 12 (2.3%) |
| *Height* | 8 (1.5%) |

**Table 1.** Frequency and percentage of missing values per variable (N = 528). Percentages calculated using the total sample size.

Missingness was predominantly confined to single variables, with minimal overlap across characteristics (Supplementary Figure 1), suggesting isolated recording omissions rather than systematic loss of patient records. To assess potential bias, missingness was examined for association with screening uptake. There was no evidence that missingness was associated with screening decision for alcohol intake ($\chi^2(1)$, $p > 0.99$), height (Fisher's exact test, $p = 0.50$), or employment status ($\chi^2(1)$, $p = 0.39$). Missing alcohol intake and height did not differ significantly across clinical teams ($p = 0.70$ and $p = 0.35$, respectively). However, all missing employment status values were observed within DN01 ($\chi^2(11)$, $p < 0.001$), suggesting a difference in recording practice within that team rather than missing information at the patient level. Alcohol intake may have been missing due to patient reluctance to disclose consumption or inconsistent documentation during consultations. Height may not have been routinely measured at the point of screening, particularly if self-reported.

Given the low proportion of missing data (<10% for all variables) and the absence of association with screening decision, complete-case analysis was considered unlikely to materially bias results. Analyses involving employment status were interpreted with caution due to its structural absence in one dataset.

## 2. Assessment of Between-Team Heterogeneity

To assess whether patient populations were comparable across the 12 clinical teams, baseline demographic and clinical characteristics were compared using one-way ANOVA (or Kruskal–Wallis tests where appropriate) for continuous variables and chi-square tests for categorical variables.

### 2.1 Baseline Patient Characteristics

Age, weight, alcohol intake and number of dependents did not differ significantly across clinical teams (all $p > 0.05$), suggesting broadly comparable patient populations. Employment status distributions were also similar across teams when excluding DN01, where employment status was not recorded.

Height showed statistically significant heterogeneity (ANOVA, $p < 0.001$), driven primarily by DN012, which had a higher mean height (~1.75 m) than other teams (~1.58–1.61 m). Review of the original data confirmed correct unit conversion, indicating this difference reflects genuine demographic variation rather than a data processing issue. As height was not a key variable of interest in screening uptake, this difference was not considered likely to affect subsequent analyses.

### 2.2 Screening Uptake Across Clinical Teams

Screening acceptance ranged from 27.8% (DN04) to 75.0% (DN08) (Figure 1). Sample sizes differed substantially between teams (range: 12–102 patients), with smaller teams exhibiting wider confidence intervals and therefore greater statistical uncertainty. Although there was statistical evidence of variation in screening uptake between teams ($\chi^2(11)$, $p = 0.045$), the strength of association was small (Cramér's $V = 0.13$). Confidence intervals overlapped across most teams, indicating that the observed differences are likely

modest in practical terms. Overall, patient populations were sufficiently similar to support pooled multivariable modelling, with adjustment for clinical team.

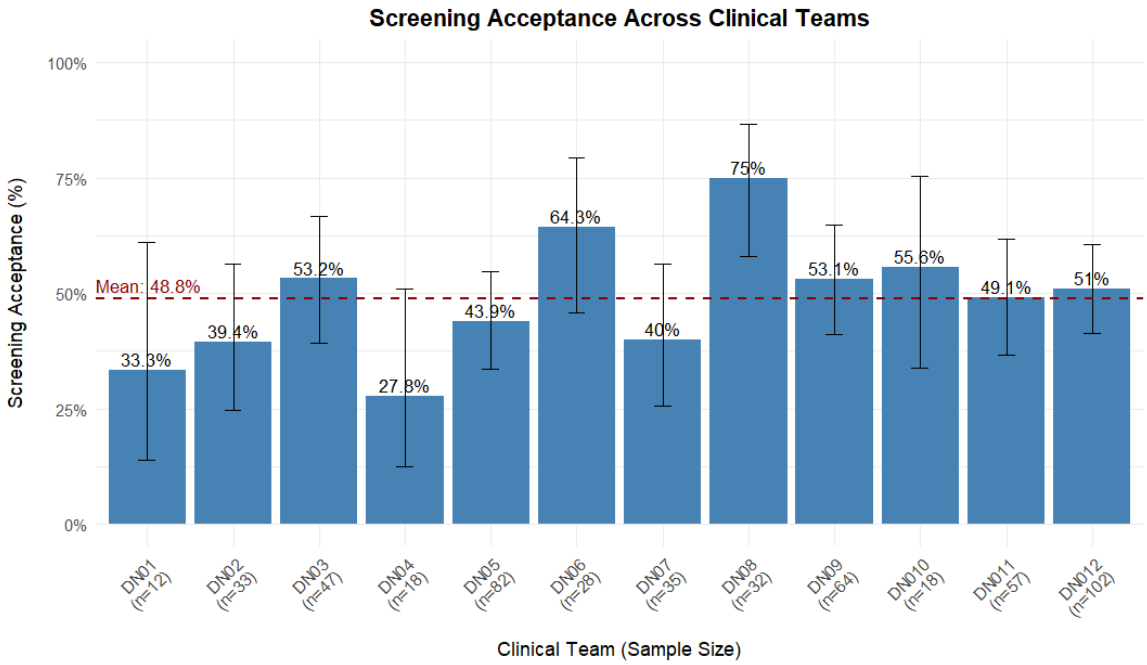


**Figure 1. Screening acceptance across clinical teams.** Bars represent the proportion of patients who accepted breast cancer screening within each clinical team. Error bars indicate 95% binomial confidence intervals. The dashed red line represents the overall mean screening acceptance across all teams (48.8%). Team sample sizes are shown below each label.

## 3. Association Between Patient Characteristics and Screening Uptake

To examine whether patient characteristics were associated with screening attendance, unadjusted comparisons were conducted between women who attended screening and those who did not. Continuous variables were compared using t-tests (or Wilcoxon rank-sum tests where appropriate), and categorical variables were compared using chi-squared tests. Independent predictors of screening uptake were then assessed using a multivariable logistic regression model. Logistic regression was selected because screening attendance is a binary outcome (Yes/No), allowing simultaneous adjustment for multiple factors. Logistic regression assumes independent observations and a linear relationship between continuous predictors and the log-odds of screening; these assumptions were considered reasonable in this setting.

In unadjusted analyses, women who attended screening were older on average than those who did not (60.6 vs 55.1 years, $p < 0.001$). Screening uptake also differed according to employment status, number of dependents, and clinical team (all $p < 0.05$), whereas alcohol intake was not associated with attendance.

In the adjusted analysis (Table 2; Supplementary Figure 2), age remained positively associated with screening uptake, with each additional year associated with approximately a 4% increase in the odds of attendance (OR 1.04, 95% CI 1.02–1.06). Compared with women in full-time employment, those working part-time had approximately 70% lower odds of attending screening (OR 0.27, 95% CI 0.12–0.57), whereas retired or unemployed women had more than five times the odds of attending (OR 5.30, 95% CI 3.26–8.83). Each additional dependent was associated with approximately a 65% reduction in the odds of screening (OR 0.34, 95% CI 0.26–0.43). Alcohol intake was not independently associated with screening.

After adjustment for patient characteristics, most differences between clinical teams were attenuated. However, patients registered with DN08 had higher odds of screening attendance compared with the reference team (DN01) (OR 4.62, 95% CI 1.24–18.90). No other clinical team showed a statistically significant

independent association. Inclusion of weight and height in a secondary model introduced statistical collinearity (i.e. strong correlation between related variables), which led to unstable estimates. This was confirmed using standard diagnostic measures. As neither variable was independently associated with screening uptake, and their inclusion did not materially change the main findings, the simpler primary model was retained.

| ***Factor*** | ***Adjusted *OR (95% CI)*** | ***p-value*** | ***Clinical interpretation*** |
|---|---|---|---|
| *Age (per year)* | 1.04 (1.02–1.06) | <0.001 | Older women were slightly more likely to attend screening |
| *Alcohol intake (per unit)* | 1.01 (0.99–1.04) | 0.39 | No clear evidence of an association |
| *Part-time vs Employed* | 0.27 (0.12–0.57) | <0.001 | Part-time workers had substantially lower screening uptake |
| *Retired/Unemployed vs Employed* | 5.30 (3.26–8.83) | <0.001 | Retired or unemployed women had markedly higher uptake |
| *Number of dependents* | 0.34 (0.26–0.43) | <0.001 | Greater family responsibilities were associated with lower uptake |
| *DN08 vs DN01* | 4.62 (1.24–18.90) | 0.027 | Screening uptake appeared higher in DN08 |
| *Other clinical teams* | **CIs cross 1 | >0.05 | No strong evidence of difference vs DN01 |

* OR = Odds Ratio ** CIs = Confidence Intervals

**Table 2. Factors associated with breast cancer screening attendance (primary model).** Adjusted odds ratios (ORs) and 95% confidence intervals from the primary regression model examining characteristics associated with screening attendance. The analysis accounts for age, alcohol intake, employment status, number of dependents, and clinical team. DN01 and full-time employment were used as comparison groups. An odds ratio above 1 indicates a higher likelihood of attending screening, whereas an odds ratio below 1 indicates a lower likelihood.

## 4. Clinical Implications and Limitations

The observed patterns suggest that barriers to breast cancer screening are likely to be practical and social rather than clinical. Differences in participation appear consistent with competing responsibilities and time constraints, indicating that accessibility may play a central role in screening uptake. From a service delivery perspective, strategies such as extended clinic hours, workplace engagement initiatives, or targeted outreach may help improve attendance among groups less likely to participate. The limited independent effect of most clinical teams after adjustment suggests that variation in uptake is largely attributable to differences in patient populations rather than differences in team performance.

Several limitations should be considered. As this was an observational analysis, the results cannot prove cause and effect. Some relevant information, such as income or education, was not available and may help explain differences in attendance. Analyses were based on complete cases, which may introduce bias if missing data were not entirely random. Additionally, relatively small sample sizes within some teams may have reduced the precision of certain estimates. Overall, improving screening uptake may require system-level approaches that address structural and social barriers rather than focusing solely on clinical factors.